

BJP TECHNOLOGIES

(T) LIMITED

SESSION 14

GOOGLE ANALYTICS 4 INTEGRATION

Date: 27 May 2026
Project: bjptechnologies.co.tz
Phase 6 — Polish & Launch

1. Overview

This session integrated **Google Analytics 4 (GA4)** into the BJP Technologies website end to end, following the GA4 Integration Playbook. The work shipped in two independent phases: a client-side tracking tag that collects visitor data, and an in-admin dashboard that reads that data back through the GA4 Data API and renders it natively inside the Django admin.

Two GA4 identifiers are used and must never be confused: the **Measurement ID** (G-C9L66H4VE4) drives the client-side tag, while the numeric **Property ID** (538536674) is used by the server-side Data API for the dashboard.

2. Phase 1 — The Tracking Tag

The gtag.js tag is injected on every public page and is fully controllable from the admin. It is emitted only when it is safe and wanted — never in local development.

Component	Behaviour
Measurement ID storage	Editable in admin via SiteSettings; seeded on first deploy
Master kill switch	ga_enabled toggle disables tracking site-wide
DEBUG suppression	Tag never renders while DEBUG=True (no polluting prod stats)
Conversion event	contact_submit fires on the contact success page
Admin entry	Analytics > Google Analytics (editable Measurement ID + switch)

Emit condition: not DEBUG and ga_enabled and bool(ga_measurement_id)

Status: Merged to develop via PR #62. Verified live in GA4 Realtime.

3. Phase 2 — The In-Admin Dashboard

A new apps/analytics app surfaces GA4 traffic directly inside the Unfold admin — same sidebar, breadcrumbs, and dark mode as the rest of the panel. All Data API calls pass through one cached gateway to stay within the ~25,000 token/day quota.

Dashboard Card	Data Shown	Cache TTL
Realtime active users	Users on the site right now	60s
Traffic KPIs	Users + page views (today / 7d / 30d)	30 min
Top pages	Most-viewed pages (7 days)	1 hr
Traffic sources	Where visitors came from (7 days)	1 hr
Top countries	Visitor countries (30 days)	1 hr
Device split	Mobile / desktop / tablet (30 days)	1 hr
Conversions	contact_submit count (30 days)	30 min

Authentication: OAuth user-token (the GA4-admin account), not a service account — GA4 will not attach a service account to a property owned by a personal Gmail. A refresh token was captured once via the ga_capture_token command and stored in environment variables; ga_verify confirmed it works.

4. Technical Traps Handled

Trap	Resolution
Measurement ID vs Property ID confusion	Documented and separated: tag uses G-, Data API uses the numeric ID
Every KPI shows 0 (reading totals[])	Read headline numbers from rows[0]; totals[] is empty without aggregations
Unfold's narrow Tailwind bundle drops utility classes	Dashboard ships its own scoped .ga-dash raw CSS instead of utilities
Dashboard page rendered with no admin chrome	Virtual model + changelist_view merging admin_site.each_context()
OAuth refresh token expires after 7 days	Consent screen published to 'In production' (not Testing)

5. Adaptations to This Codebase

Playbook assumed	This repo uses	Adaptation
UUID singleton PK	Integer pk=1	Seed migration targets pk=1
django-enviro env()	python-dotenv + os.environ	os.environ.get(...)
Inline success render	FormView -> redirect	Event fired on success view
Many conversion events	One contact form	Single event: contact_submit

6. Quality & Verification

- **86 / 86** tests passing (19 new: 6 Phase 1, 13 Phase 2)
- ruff and black clean on all new files
- Live API confirmed: ga_verify returns 'Auth works'; dashboard context builds with no errors
- Dashboard rendered and screenshotted in admin — chrome intact, brand gradients applied

7. Current State & Next Steps

Item	Status	Notes
Phase 1 — tracking tag	Complete	Merged to develop (PR #62), live
Phase 2 — dashboard	Complete (code)	Branch pushed; pending deploy
Server GA_* env vars	Pending	Add 4 vars in cPanel Python App
Dashboard data	Awaiting data	Zeros until ~48h of collection
GA4 Key Event	Pending	Mark contact_submit as Key Event
OAuth 'In production'	To confirm	Prevents 7-day token expiry

Why the dashboard currently shows zeros: GA4 standard reports (users, page views, top pages, etc.) lag **24–48 hours** on a freshly created property — this is normal Google processing latency, not a fault. Authentication is confirmed working. Realtime confirms data arrival immediately; the historical cards will populate automatically as data accumulates.

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